

8. Compound **A** contains only carbon, hydrogen and an unknown halogen.

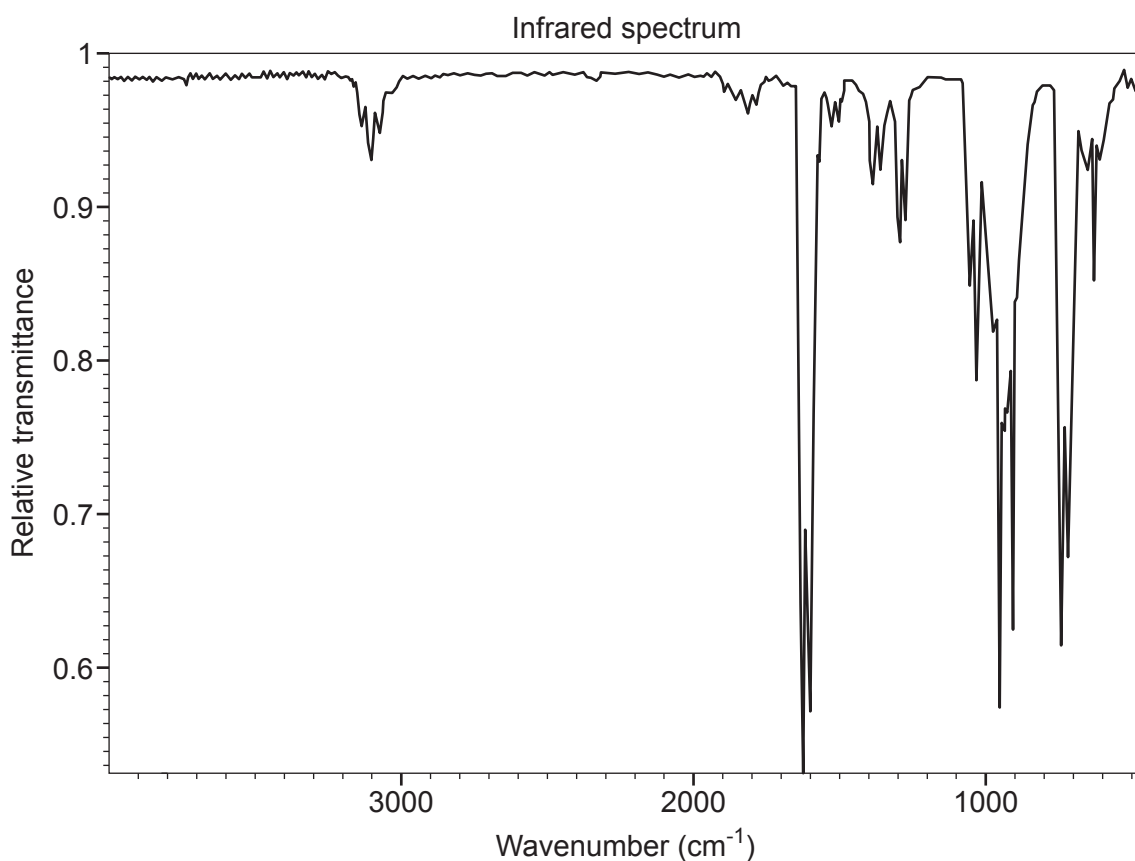
Refluxing compound **A** in aqueous sodium hydroxide followed by the addition of nitric acid and aqueous silver nitrate produces a white precipitate.

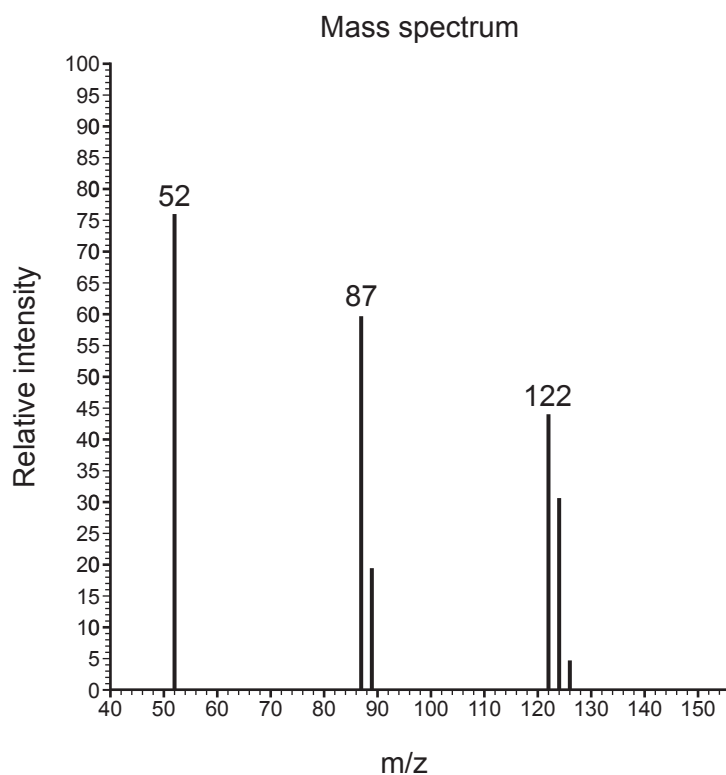
Elemental analysis of compound **A** indicates it contains 39.02% carbon and 3.25% hydrogen by mass.

When bromine is added to compound **A**, 123 g of compound **A** reacts with 320 g of bromine.

The ^1H NMR spectrum of compound **A** consists of only one peak. The ^{13}C NMR spectrum of compound **A** consists of two peaks.

The infrared spectrum and simplified mass spectrum are shown below and overpage.





Use **all** of the data provided to determine the identity of compound **A**. Explain your answer using information from all of the data sources provided.

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